Question:

An investigator is studying mechanisms of urea excretion in humans. During the experiment, a healthy male volunteer receives a continuous infusion of para-aminohippurate (PAH) to achieve a PAH plasma concentration of 0.01 mg/mL. A volume of 1.0 L of urine is collected over a period of 10 hours; the urine flow rate is 1.66 mL/min. The urinary concentration of PAH is measured to be 3.74 mg/mL and his serum concentration of urea is 0.2 mg/mL. Assuming a normal filtration fraction of 20%, which of the following best estimates the filtered load of urea in this patient?Answer Choices:A. 25 mg/minB. 124 mg/minC. 7 mg/minD. 166 mg/min

Reasoning + Answer:

<think>considering the filtration fraction and urinary excretion rates to estimate the urea filtered load</think><answer>124 mg/min</answer>